Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Brainy Barbies
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4
Project / Solution Name	Future Path
Name (Leader)	Law Jing Xuan
Email & Phone (Leader)	lawjingxuan1234@gmail.com (016-2299315)
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Due to the rapid development of AI, recent graduates are unable to keep up with on-demand skills as there is a gap in what they learn in universities compared with what they can actually apply in workplace settings, resulting in reviews made by MOHE to keep pace offered courses with AI and industry needs (Ministry of Higher Education Malaysia (MOHE), 2026). Therefore, it is essential to raise awareness among fresh graduates on current and potential future job market trends for better decision-making in terms of their learning priorities and career selection. This way, they can find space and survive in the job market and thrive throughout their careers as our solution tackles this issue.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed solution, FuturePath AI, uses artificial intelligence to help fresh graduates prepare for changes in the job market by identifying suitable career opportunities, predicting emerging workforce trends, and highlighting skills that may face increasing automation risk.

Users can upload their CVs or manually enter their existing skills and experience. The application analyses this information to identify skills explicitly demonstrated by the candidate and suggest potentially relevant skills that lack sufficient evidence in the provided profile. The analysis uses AI models and relevant datasets to build a clearer picture of each candidate's capabilities.

Based on the candidate's profile and available job-market data, FuturePath AI aims to recommend suitable career pathways and personalised upskilling opportunities. It also maps relevant skills and career pathways

to the United Nations Sustainable Development Goals (SDGs), helping users understand how their professional development may contribute to broader societal goals.

The application uses a frontend-backend architecture, with backend services communicating with the Google Gemini API for CV and skills analysis. The project also intends to leverage public datasets from Kaggle to support workforce-related analysis. Through the SDC Phase 1 workshops, we aim to apply skills in generative AI API integration, data handling, backend development, and cloud-based application deployment to build an accessible, AI-powered career planning tool for fresh graduates.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Our project is a web-based Skills and Career Recommendation application based on CV extraction. It helps users extract structured information from their CVs, analyse their skills and recommend job opportunities for them with ranking.

- **Replit Core:** Used to develop and deploy the web application through the SDC sponsorship.
- **React and TypeScript:** Used to build the frontend interface
- **Node.js and TypeScript:** Used for backend API handling, document processing, communication with AI services, and response validation.
- **Qwen3-VL-4B-Instruct:** A vision-language model used for extracting structured candidate information from CV content.
- **RunPod Serverless:** Used to host and run Qwen3-VL-4B-Instruct through an A40 GPU and a vLLM-based inference service, supported by the SDC GPU sponsorship.
- **Google Gemini API:** Used to analyse extracted candidate skills and experience and generate career-related insights and recommendations.
- **Hugging Face:** Used to access AI model resources and support model development and training workflows.
- **Google Collab:** Used for notebook-based AI development and model training experiments.
- **Kaggle:** Used to discover publicly available datasets for training data preparation and demonstrations.
- **Document Processing:** PDF text extraction and page rendering, with support for JPG and PNG image uploads.
- **Secrets Management:** Replit Secrets for securely storing API credentials.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We started by identifying core frustration for our target users (fresh graduates), then we defined our problem statement to focus entirely on 1 type of user only for now. Our prototype is then built using React and TypeScript for frontend interface and Node.js and TypeScript for secure backend services, while Google Gemini API key is being used for frontend to call backend services. Data is handled using the database within Replit Core and the data used to run our AI model comes from Kaggle.

Instead of a hidden box algorithm, our backend calculates a transparent, human-readable skills-gap breakdown and match score using the skills data extracted from fictional CVs and computes structural gaps against real-time emerging roles. This allows the users to understand the result well, at the same time making this solution user-friendly.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4 - Relevant skills for decent jobs SDG 8 – Devent work and economic growth
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	62% of university students believed they were well-prepared for the workforce while only 32.3% of employers agree that local universities adequately prepare graduates for workplace demands as AI reshapes entry-level roles (REHDA Institute, 2026).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% accuracy of skills and roles matched to the candidate's profile based on data extracted from their CVs.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 10% improvement in computing roles and skills matched to candidates.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

No

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Kaggle	**Dataset**	Kaggle dataset license

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

No

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Charlotte]

24/10/2026, Saturday